

Center-slice noise in cone-beam MBIR: mechanism, remedy, and validation

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Abstract

Circular cone-beam MBIR reconstructions computed with diagonally preconditioned, line-searched subset iterations can show a band of noisy, slowly converging slices at the axial center of the volume. We show that this artifact is a preconditioning failure, not a data or modeling problem: the update step is exact Newton on a separable quadratic model, and for the single-slice DC directions near the plane of the source the model understates the true curvature by a large, geometry-determined factor. The exact line search couples the step size to typical directions, so these directions are overstepped and fail to contract; the visible artifact is the preserved and continually re-excited error along them — a property of the iteration path, while the minimizer itself is artifact-free. We derive the curvature-gain profile from scanner geometry, measure it across datasets, and show that whether a given scan actually develops the artifact is governed by the realized drive, not the gain. The proposed remedy is a rank-one, per-slice damping of each subset update’s in-plane mean component, with a profile computed from the measured gain and the reconstruction’s own curvature scales, and an automatic decide-once trigger that disables the damping on undriven data. Since it is a preconditioner, the damping leaves the minimizer unchanged. On available data, this approach removes the artifact at matched-or-better data fit with negligible cost. [TODO: finalize numbers cited here after Section 7 is written]

1 Introduction

[TODO: plume figure: axial noise profile and/or slice montage, undamped vs damped, before this paragraph or beside it]

At moderate-to-high sharpness settings, circular cone-beam MBIR reconstructions can exhibit a narrow band of slices near the axial center of the volume that is visibly noisier than the rest and converges more slowly — up to an order of magnitude above the local noise level on the worst dataset we have measured, decaying only slowly with additional iterations (Section 7). The band’s location is not a property of the object: it sits at the plane of the x-ray source. This report explains the artifact, proposes a remedy, and documents where the remedy does and does not apply.

Briefly, the artifact arises from a mismatch between the true Hessian of the data term and the diagonal approximation used as its preconditioner; this mismatch is severe only for slices near the source plane (the center slices). In each iteration, the gradient is divided per-voxel by the diagonal curvature and scaled by a single line-searched step size. That update is exactly

Newton’s method on a separable model of the cost, and the separable model is exact for single-voxel updates (ICD). It is systematically wrong for one family of directions: the DC component of a single slice near the source plane, raising all of that slice’s voxels equally. For that family, cone-beam geometry makes the true curvature exceed the separable model’s curvature by a large factor. Every pair of voxels in the source-plane slice lands on the same detector row in every view, and in the views where these voxels also lie on the same ray from the source, they land on the same detector element; accumulating these coincidences through a full rotation leads to large off-diagonal Hessian couplings across the entire source-plane slice. In contrast, on a slice far from the source plane, two voxels land on different rows in nearly every view unless they are close neighbors, so only local couplings survive and the diagonal is an adequate model (both regimes are measured directly in Figure 2). The line search applies one step size to the whole update, so it is calibrated by the many directions the separable model handles well rather than the few it does not. The DC components of slices near the source plane are therefore overstepped on every subset update, and they accumulate a persistent, noise-fed error. The band of slices carrying those components is the artifact.

The report makes six claims.

1. **Mechanism.** The update step is exact Newton on a separable model. This update step overrepresents the single-slice DC components in slices for which the true forward model curvature is greater than the separable curvature. We use this ratio to define the scale-invariant function $g(\Delta k)$. Any direction with $g > 1$ is overstepped by exactly that factor (Section 3).
2. **Geometry.** Circular cone-beam geometry makes g peak at the source plane ($\Delta k = 0$) and decay away from it, through magnification-independent row placement at $z = z_c$; the natural axial variable is the offset in projected detector rows, $\zeta = W_{pr} \Delta k$ (Sections 4 and 5).
3. **Instability.** The exact line search sets $\alpha = 1/R(\tilde{v})$, tying the step size to the directions that dominate the update; slices with $g \geq 2R(\tilde{v})$ are then non-contractive, and the top of the operator’s spectrum is measured to be essentially the slice-DC family itself (Sections 3 and 5).
4. **Artifact character.** The artifact is a property of the iteration *path*, not of the minimizer: preserved initialization error plus continually re-injected subset fluctuation along the non-contracting directions. Step-side remedies are therefore the right class, and cannot bias the answer (Section 3).
5. **Severity.** Whether a given scan develops the artifact is governed by the realized *drive* — the residual’s excitation of the family — not by the gain alone: the gain profile is similar across scans, while immunity is common. This is what makes an automatic trigger possible (Section 5).
6. **Remedy.** Rank-one damping of each subset update’s in-plane mean component, with the per-slice profile $s = (\tilde{f}_h + \bar{p}_h)/(g\tilde{f}_h + \bar{p}_h)$ floored at a measured knee, restores contraction while leaving per-voxel structure at full speed and provably preserving the minimizer; a decide-once drive trigger stands the damping down on undriven data. Measured: the artifact is removed at matched-or-better data fit, with no axial-resolution cost at the plane and negligible wall-clock cost (Sections 6 and 7).

Section 2 fixes the setting and notation. Section 3 develops the analysis with no reference to geometry. Section 4 supplies the cone-beam geometry. Section 5 defines the measurements and reports them. Section 6 derives the remedy and its safety properties, and Section 7 reports the validation. Section 8 records limitations and directions for improvement.

2 Setting

The reconstruction estimates a volume $x \in \mathbb{R}^{n_r \times n_c \times n_s}$ (rows $\times$ columns $\times$ slices; slices indexed $k = 0, \dots, n_s - 1$ along the rotation axis z) from a cone-beam sinogram y by minimizing the MAP cost

$$c(x) = \frac{1}{2\sigma_y^2} \|y - Ax\|_W^2 + \rho(x), \quad (1)$$

where A is the cone-beam forward projector, W the (diagonal) sinogram weights, σ_y the noise scale, and ρ the qGGMRF prior. The VCD (vectorized coordinate descent) iteration updates one *subset* of in-plane pixel positions at a time: for each pixel j in the subset it forms the cost gradient and divides by the **per-voxel curvature diagonal**

$$d_j(k) = f_h(j, k) + p_h(j, k), \quad f_h(j, k) = \frac{1}{\sigma_y^2} [A^\top W A]_{(j,k),(j,k)}, \quad (2)$$

i.e. the data term's Hessian diagonal plus the prior's local curvature, and then applies one global step size α chosen by exact line search over the subset. The resulting per-slice update array for the subset is written $\delta(\cdot, k)$.

This diagonal-plus-line-search step is accurate for updates that look like typical isolated voxels. It is systematically wrong for one particular family of updates, described next.

3 When diagonal preconditioning overshoots

The direction, and its true curvature. Fix a slice k . Let S_k denote the set of its in-support voxels, $N_k = |S_k|$, and $\mathbf{1}_k$ the indicator image of S_k . Define

$$u_k = \frac{1}{\sqrt{N_k}} \mathbf{1}_k, \quad (3)$$

the **normalized single-slice DC direction**: the image that raises every in-support voxel of slice k equally and touches nothing else ($\|u_k\|_2 = 1$). Write $f(x) = \frac{1}{2\sigma_y^2} \|y - Ax\|_W^2$ for the data term, with constant Hessian $H = \nabla^2 f = \frac{1}{\sigma_y^2} A^\top W A$ and diagonal part $D = \text{diag}(H)$, so $D_{(j,k),(j,k)} = f_h(j, k)$ from Eq. (2). (This section works with the data term alone; the prior's share of the curvature is folded in in Section 6.) The curvature of the data term along u_k is the second derivative of the one-variable restriction $f_k(r) := f(x + r u_k)$:

$$f_k''(0) = u_k^\top H u_k = \frac{1}{\sigma_y^2 N_k} \|W^{1/2} A \mathbf{1}_k\|^2 \quad (4)$$

— computable by one weighted forward projection of a slice-indicator image.

The separable-model curvature. The step $\delta = -D^{-1} \nabla f(x)$ is not an ad-hoc scaling: it is the exact minimizer of the separable quadratic model

$$m(x + \delta) = f(x) + \nabla f(x)^\top \delta + \frac{1}{2} \delta^\top D \delta, \quad (5)$$

i.e. Newton’s method applied to a separable model of f whose Hessian is D instead of H . (“Separable model” throughout refers to this surrogate, distinct from the forward model A and the prior model ρ .) Restricting the separable model to the same line, $m_k(r) := m(x + r u_k)$ has the same slope as f_k at $r = 0$ but curvature

$$m_k''(0) = u_k^\top D u_k = \frac{1}{N_k} \sum_{j \in S_k} H_{jj} = \bar{f}_h(k), \quad (6)$$

the mean of the Hessian diagonal over slice k , which we abbreviate $\bar{f}_h(k)$. Define the **gain** as the ratio of the true to the separable-model curvature:

$$g(k) := \frac{f_k''(0)}{m_k''(0)} = \frac{u_k^\top H u_k}{u_k^\top D u_k} = 1 + \frac{u_k^\top (H - D) u_k}{u_k^\top D u_k}. \quad (7)$$

The numerator of the last term collects exactly the pairwise couplings among the slice’s voxels: $u_k^\top (H - D) u_k = \frac{1}{N_k} \sum_{i,j \in S_k, i \neq j} H_{ij}$. Along the line specified by u_k , the separable model’s one-dimensional minimizer sits at $r_m = -f_k'(0)/m_k''(0)$ while the true one is $r^* = -f_k'(0)/f_k''(0)$, so $r_m = g(k) r^*$. In the case that $g > 1$, the separable model’s target overshoots the true one by exactly that factor. For an axis direction e_j (an isolated voxel), $g = H_{jj}/D_{jj} = 1$ identically: the separable model is *exact* for the directions it is built from, and $g(k)$ measures how far a single-slice DC departs from this single-voxel case.

The invariant meaning of g . A raw directional second derivative such as $f_k''(0)$ carries no intrinsic meaning by itself: it scales with the normalization of the direction ($u_k \rightarrow c u_k$ multiplies it by c^2), and even at unit Euclidean norm it depends on the arbitrary per-voxel units. Concretely, rescale the voxels by any positive diagonal matrix Λ :

$$\tilde{x} := \Lambda x, \quad \tilde{f}(\tilde{x}) := f(\Lambda^{-1} \tilde{x}). \quad (8)$$

The transformed objective has Hessian $\tilde{H} = \nabla^2 \tilde{f} = \Lambda^{-1} H \Lambda^{-1}$ with diagonal part $\tilde{D} = \Lambda^{-1} D \Lambda^{-1}$, and the perturbation that raises slice k ’s in-support voxels equally is now the vector $\tilde{u}_k = \Lambda u_k$. The Euclidean-normalized curvature changes with Λ ,

$$\frac{\tilde{u}_k^\top \tilde{H} \tilde{u}_k}{\|\tilde{u}_k\|_2^2} = \frac{f_k''(0)}{\|\Lambda u_k\|_2^2},$$

but the ratio g does not: its numerator and denominator are evaluated on the same vector and transform identically ($\tilde{u}_k^\top \tilde{H} \tilde{u}_k = u_k^\top H u_k$ and $\tilde{u}_k^\top \tilde{D} \tilde{u}_k = u_k^\top D u_k$), so g is a property of the objective and the perturbation alone, independent of how the voxels happen to be scaled.

Among all such rescalings there is a canonical one — the one the algorithm itself defines: $\Lambda = D^{1/2}$, so that from here on

$$\tilde{x} = D^{1/2} x, \quad \tilde{f}(\tilde{x}) = f(D^{-1/2} \tilde{x}), \quad \tilde{H} = D^{-1/2} H D^{-1/2}. \quad (9)$$

Since $\nabla \tilde{f}(\tilde{x}) = D^{-1/2} \nabla f(x)$, plain (unpreconditioned) gradient descent in $\tilde{x}$ maps back to exactly the preconditioned step in x :

$$\tilde{x}^+ = \tilde{x} - \alpha \nabla \tilde{f}(\tilde{x}) \iff x^+ = x - \alpha D^{-1} \nabla f(x), \quad (10)$$

and $\tilde{H}$ has *unit diagonal* ($\tilde{D} = I$). In these coordinates every axis direction has curvature exactly 1 (recovering the e_j remark above), and the normalized transformed direction $\tilde{u}_k/\|\tilde{u}_k\|_2 = D^{1/2}u_k/\|D^{1/2}u_k\|_2$ has ordinary Euclidean-normalized curvature

$$\frac{\tilde{u}_k^\top \tilde{H} \tilde{u}_k}{\|\tilde{u}_k\|_2^2} = \frac{u_k^\top H u_k}{u_k^\top D u_k} = g(k). \quad (11)$$

Equivalently, in metric language: the separable model (5) is the *isotropic* quadratic $\frac{1}{2}\|\delta\|_D^2$ in the inner product $\langle a, b \rangle_D = a^\top D b$ — by construction it has curvature 1 in every direction of that metric — while the objective’s actual curvature in the same metric along u_k is $u_k^\top H u_k / \|u_k\|_D^2 = g(k)$. Diagonal preconditioning is exactly the bet that the objective is nearly isotropic in its own metric; $g(k)$ is the anisotropy along u_k that survives.

Why large g breaks the iteration: the spectral statement. The iteration we analyze is the repeated application of the whitened step, Eq. (10) — the full-volume version of the update of Section 2, restricted to the data term (the subset and prior refinements are noted at the end of the paragraph and in Section 6). Because $\tilde{f}$ is quadratic, its gradient is linear in the distance from the minimizer $\tilde{x}^*$: $\nabla \tilde{f}(\tilde{x}) = \tilde{H}(\tilde{x} - \tilde{x}^*)$. Writing $\tilde{e} := \tilde{x} - \tilde{x}^*$ for the error and subtracting $\tilde{x}^*$ from both sides of Eq. (10),

$$\tilde{e}^+ = \tilde{e} - \alpha \tilde{H} \tilde{e} = (I - \alpha \tilde{H}) \tilde{e} : \quad (12)$$

each step multiplies the error by the fixed matrix $I - \alpha \tilde{H}$. $\tilde{H}$ is symmetric positive semidefinite, so it has an orthonormal eigenbasis with nonnegative eigenvalues; expanding $\tilde{e}$ in that basis, the coefficient on an eigenvector with eigenvalue λ is multiplied by $(1 - \alpha\lambda)$ at every step, so that coefficient shrinks if and only if $0 < \alpha\lambda < 2$.

By the criterion just derived, every component of the error contracts precisely when $\alpha\lambda < 2$ for every eigenvalue λ of $\tilde{H}$ — equivalently, when $\alpha \lambda_{\max}(\tilde{H}) < 2$. The convergence question thus reduces to a single number, $\lambda_{\max}(\tilde{H})$. We cannot compute it exactly: $\tilde{H}$ is far too large to form, and it can only be applied to chosen vectors, at the cost of one forward and one back projection per application. But to establish *failure* an exact value is not needed: a lower bound on $\lambda_{\max}$ suffices, and lower bounds are cheap. For a symmetric matrix the Rayleigh quotient $R(v) = (v^\top \tilde{H} v) / (v^\top v)$ satisfies $\lambda_{\max}(\tilde{H}) = \max_{v \neq 0} R(v)$, so its value at *any* single vector is a lower bound on $\lambda_{\max}$. The direction of the bound is what makes the argument cheap: each evaluation of R can only push the certified maximum up, so no collection of test vectors could supply the *upper* bound on $\lambda_{\max}$ that a proof of convergence would need — but a single well-chosen vector can certify failure. Since $R(D^{1/2}u) = (u^\top H u) / (u^\top D u)$, the gain of Eq. (7) is exactly such a value: the single-slice DC directions certify, for every slice k ,

$$\lambda_{\max}(\tilde{H}) \geq g(k). \quad (13)$$

By itself a large eigenvalue is harmless if α is small enough: shrinking α below $2/\lambda_{\max}$ would make every coefficient contract. But the algorithm does not choose α that way. The update direction is $\tilde{v} := -\nabla \tilde{f}(\tilde{x})$, and the line search minimizes the quadratic $\alpha \mapsto \tilde{f}(\tilde{x} + \alpha \tilde{v})$ exactly; setting its derivative to zero gives

$$\alpha = \frac{\tilde{v}^\top \tilde{v}}{\tilde{v}^\top \tilde{H} \tilde{v}} = \frac{1}{R(\tilde{v})} \quad (14)$$

(the same identity holds per subset for the subset-restricted quantities). The step size is therefore matched to the curvature of the direction the iteration is currently moving along — not to the largest curvature present. Combining Eqs. (13) and (14): whenever some slice satisfies

$$g(k) \geq 2R(\tilde{v}), \quad (15)$$

we have $\alpha \lambda_{\max} \geq \alpha g(k) \geq 2$, so $I - \alpha \tilde{H}$ has an eigendirection whose coefficient satisfies $|1 - \alpha \lambda| \geq 1$: some component of the error fails to shrink, and grows when the inequality is strict. In words: the error stops contracting whenever the gain along some slice’s DC direction is at least twice the Rayleigh quotient of the direction the line search is serving. How much danger this represents depends on where $R(\tilde{v})$ actually sits, and that is not something the analysis can settle by itself: $R(\tilde{v})$ is a property of the residual the iteration happens to be working on, not of the operator. Two reference levels bracket it. An isolated voxel — the kind of direction the separable model is built from — has whitened curvature exactly 1; were the update composed of such directions, α would sit near 1, and Eq. (15) would then be met by any slice with $g(k) \gtrsim 2$. A residual loaded instead onto coherent directions carries their much larger curvature into $R(\tilde{v})$, and α shrinks in proportion. Nothing needs to be assumed here, because the line search reports α at every step: $R(\tilde{v}) = 1/\alpha$ is an observable of the running algorithm. Section 5 measures it, and thereby places the contraction boundary $2R(\tilde{v})$ on the same scale as the measured gain profile.

Three questions address the relevance of this analysis, with answers that are primarily empirical and supported by well-defined measurements taken up in Section 5:

1. How large is $g(k)$ for the slices near the source plane?
2. Where does the line search actually put the step size — that is, how large is $R(\tilde{v})$, and hence how far must a slice’s gain rise above the volume’s ambient level before Eq. (15) bites?
3. Is the bound (13) nearly attained by u_k ? I.e., is the single-slice DC family a strongly non-contracting eigendirection?

From non-contraction to the visible artifact. Two different objects must be distinguished here. The first is the minimizer $\hat{x}$ of the cost: it is defined by the cost alone, so nothing we do to the *step* can move it — Section 6 shows that any positive per-direction rescaling of the update leaves the fixed point of the iteration exactly at $\hat{x}$. The second is the iterate x_t the algorithm actually produces after t steps, and it is here that non-contraction does its damage: the artifact is a property of the *path*, not of the minimizer.

Consider the coefficient of the error $e_t = x_t - \hat{x}$ along the non-contracting eigendirection. In the idealized analysis above — the full-volume operator and a fixed α — that coefficient would grow without bound. The real iteration departs from the idealization in three ways, and together they convert unbounded growth into bounded non-convergence. First, α is re-chosen at every subset and fluctuates with the composition of the residual, so the per-step multiplier $(1 - \alpha \lambda)$ varies around magnitude one rather than sitting at a fixed amplifying value. Second, each step applies the operator of one randomly assigned *subset*, not the full volume, which further randomizes the multiplier. Third, each subset step differs from the full-gradient step, and that difference — driven by the uneven split of the residual, measurement noise included, across subsets — acts as a fresh perturbation with a component along the eigendirection. The coefficient therefore evolves as a stochastic recursion with per-step gain near one and persistent excitation: it neither decays nor explodes, but wanders in a band whose width is set by the excitation.

At any stopping iteration the error along the eigendirection thus has two parts: whatever the initialization placed there, which a contracting iteration would have removed and this one preserves; and the fluctuating part continually re-injected by the subset iteration. Granting the identification in question 2 above — the eigendirection concentrated on the single-slice DC components of a band of slices near the source plane — this error appears as elevated, slowly clearing noise in exactly those slices: the “plume.” The minimizer $\hat{x}$ contains no plume; the plume lives entirely in the path. That is why the remedy of Section 6 modifies the step and not the cost function (1).

4 Cone-beam geometry: why the source plane

Everything so far defines $g(k)$ and explains what a large value does; nothing yet says *which* slices should have one. That is where the scanner’s geometry enters. By Eq. (7), $g(k)$ exceeds 1 by the total pairwise coupling among the slice’s voxels, measured against the diagonal, and each coupling is explicit:

$$H_{ij} = \frac{1}{\sigma_y^2} \sum_m w_m A_{mi} A_{mj} \quad (16)$$

— a weighted inner product, over all detector measurements m in all views, of the two voxels’ projection footprints. H_{ij} is large when voxels i and j project onto the *same detector cells* in *many views*. The question “which slices have large gain” therefore reduces to a geometric one: when do two voxels of the same slice deposit weight on the same detector cell, and for which slices does that coincidence persist view after view? A detector cell is the intersection of a channel (the transaxial coordinate) and a row (the axial coordinate), so coincidence requires alignment in both coordinates at once — and the two halves of the question have completely different characters.

Channel alignment is *democratic*: a pair lines up in channel in the few views whose rays pass through both of its voxels, and this happens for every pair, in every slice, a few times per orbit. It rations *when* a pair can coincide, but it rations every slice identically, so it cannot be what makes the source plane special. What differs from slice to slice is whether, in those channel-aligned views, the pair *also* agrees in row. That is decided by the axial geometry — and it is decided completely.

Figure 1 shows the decisive construction: one view, seen side-on in the axial plane containing a channel-aligned pair — both voxels at axial offset z from the source plane, at distances $d_1 < d_2$ from the source. The ray from the source to a voxel at offset z and distance d leaves with slope z/d and meets the detector at row $\approx z m / \Delta_{\text{row}}$, where $m = D_{\text{sdd}}/d$ is that voxel’s magnification (D_{sdd} the source–detector distance): each voxel’s offset arrives at the detector magnified by its *own* m . For a source-plane pair, $z = 0$, and both rays lie inside the source plane itself: they land on the source row regardless of d_1 , d_2 , or the view. Row agreement is automatic — for every pair, in every channel-aligned view, all the way around the orbit — and the pairwise sum in Eq. (7) accumulates with nothing cancelling. For a pair at $z \neq 0$ the nearer voxel is magnified more, and the rows separate by $\approx z(m_1 - m_2)/\Delta_{\text{row}}$ rows: a gap that grows with the slice’s offset and with the pair’s in-plane separation. Once it exceeds one row, the coincidence is gone. So at offset z only pairs close enough together that $z \Delta m \lesssim \Delta_{\text{row}}$ still contribute — a neighborhood that shrinks as $|z|$ grows — and the pairwise mass in the numerator of Eq. (7) drains away.

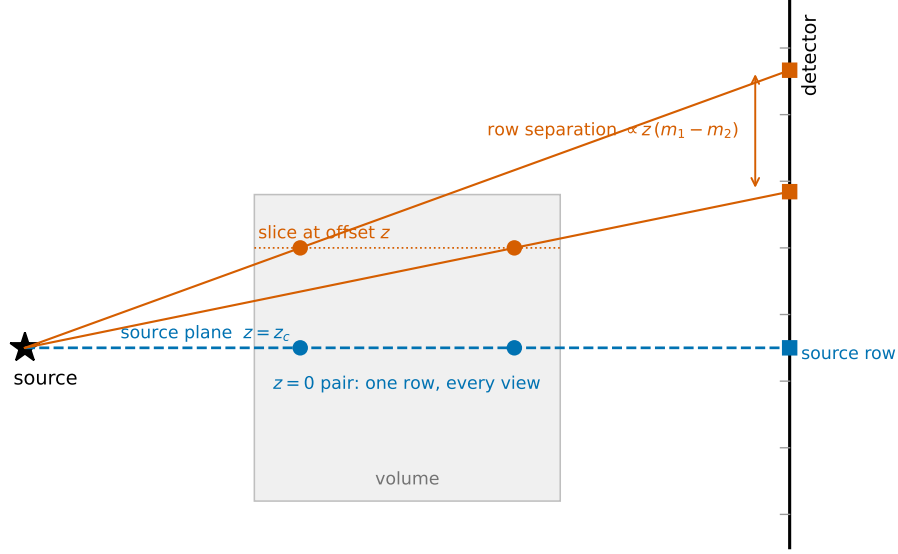


Figure 1: Why row coincidence is automatic at the source plane and fails off it. Side view of one view’s axial plane containing a channel-aligned pair. A source-plane pair (blue) lies on a ray that is contained in the source plane itself, so both voxels land on the source row — regardless of their positions and of the view. A pair at axial offset z (orange) has each voxel’s offset magnified by its *own* magnification $m = D_{\text{sdd}}/d$, so the nearer voxel lands higher and the rows separate in proportion to $z(m_1 - m_2)$.

Figure 2 shows this structure directly. Each column applies the diagonally preconditioned data operator to a single unit voxel and displays the resulting row of $D^{-1}A^TW A$ — the voxel’s couplings to the entire volume. At $\Delta k = 0$ the couplings form a band spanning the full inscribed diameter while remaining confined to one or two slices: a row of the operator that is, to a good approximation, “the whole source-plane slice.” Away from the plane the band opens into a bowtie, and the bowtie is the row-matching locus made visible: the probe at distance d_{probe} and offset z_0 couples to partners at distance d_p and offset $z_p \approx z_0 d_p / d_{\text{probe}}$ — a line through the probe with slope proportional to z_0 , joined by its mirror image from views on the opposite side of the orbit. The bowtie steepens as Δk grows, and by $\Delta k = 150$ only a compact neighborhood of couplings remains: the local floor, and the regime in which the diagonal is an adequate model of the row.

From the plane to slice coordinates. The statement just derived concerns the continuous plane $z = z_c$; everything downstream — the measured profile, the remedy — is indexed by slices, so we need the plane’s position in slice coordinates. Here z_c is the source’s axial coordinate: source and detector are fixed and the object rotates, so there is one such plane, and it meets the detector in a single row, the source row. With slice k centered at $z(k) = (k - \frac{n_s-1}{2})\Delta_z + o$, where Δ_z is the slice pitch and o the volume’s axial offset, solving $z(k_0) = z_c$ gives

$$k_0 = \frac{n_s - 1}{2} - \frac{o - z_c}{\Delta_z}, \quad \Delta k := |k - k_0|. \quad (17)$$

k_0 need not be an integer. When it falls between two slices, the plane’s coherence is shared

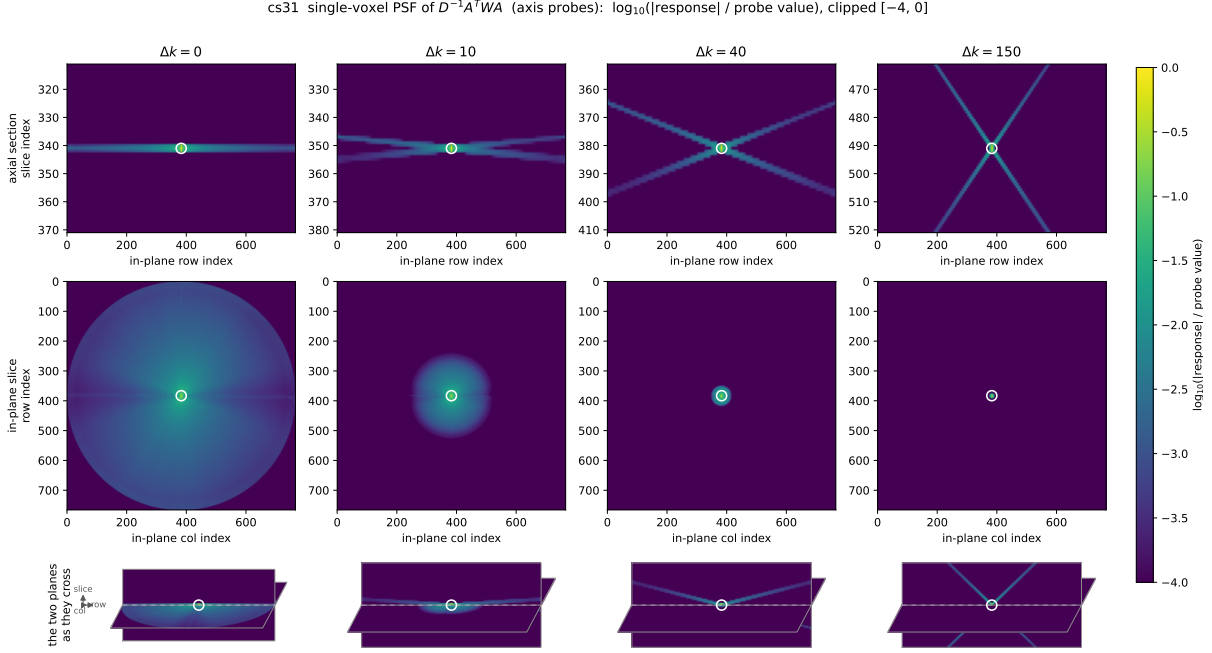


Figure 2: Measured rows of the preconditioned data operator $D^{-1}A^TWA$: the response to a single unit voxel (open circle) at $\Delta k = 0, 10, 40, 150$ on the widest-cone geometry, computed with one forward and one back projection per probe. Top: axial sections through the probe (in-plane position versus slice index, cropped to ± 30 slices). Middle: the probe’s own slice, at full extent, so that the four columns share one spatial scale. Bottom: the same two planes drawn as they sit in the volume, crossing perpendicularly along the row axis through the probe. This is a parallel projection viewed nearly along the column axis, so the axial section is seen close to face-on while the in-plane slice recedes and has its far half hidden behind the section; the slice axis is exaggerated $6\times$ so that the thin axial sheet is legible. Color is $\log_{10}$ of the response relative to the probe voxel, clipped to $[-4, 0]$.

between the straddling pair rather than concentrated in one slice — a grid-phase fact whose consequences for the remedy appear in Section 6.

The geometric prediction is therefore sharp in structure: g peaks at the source-plane slice k_0 , decays with Δk at a rate set by the cone’s divergence (a wider cone separates the rows of channel-aligned pairs faster, isolating the source plane more sharply from the bulk), and approaches its far-field level away from the plane. An order-of-magnitude account of the height and decay follows below; the actual profile is a measurement, taken up in Section 5. One structural remark completes the picture: the argument used only the row and channel alignment of voxel pairs within a slice, so it applies verbatim to any *subset* of the slice’s voxels — the pairs that remain still share rows at $z = 0$. Every VCD subset therefore sees the same coherence through its own in-plane mean, which is why the overshoot acts at every subset granularity and why the remedy of Section 6 targets each subset’s mean component.

The physical axial variable. The row argument measures axial distance in detector rows, not in slices: what destroys a pair’s overlap is its row separation. The conversion factor is the

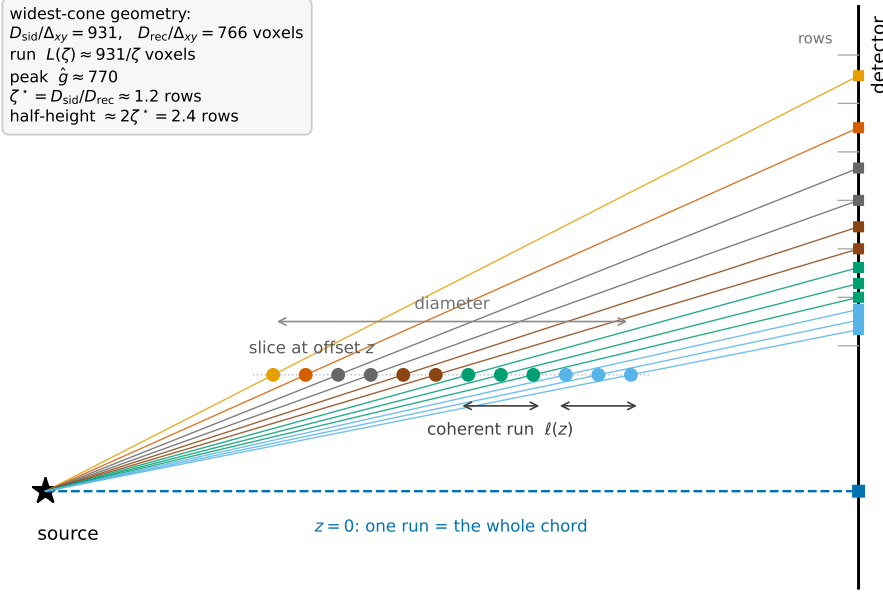


Figure 3: The coherent run, annotated with the back-of-the-envelope values for the widest-cone geometry. Voxels along one channel’s line across the reconstruction (colored by the detector row their ray reaches) land on the same row only in stretches — the coherent runs — whose length grows toward the far side of the volume and shrinks with the slice’s offset z ; at $z = 0$ (blue) the entire diameter is a single run. Contributions in the same cell are summed before squaring, while distinct runs add their squares, giving the estimate of Eq. (20) and the evaluated numbers in the box.

projected slice pitch

$$W_{pr} := \frac{\bar{m} \Delta_z}{\Delta_{\text{row}}}, \quad \zeta := W_{pr} \Delta k, \quad (18)$$

where $\bar{m}$ is the magnification at the rotation axis, so that W_{pr} is the number of detector rows one slice spans and ζ is a slice’s axial offset from the plane in projected rows. If row coherence is the governing mechanism, then profiles measured at different voxel pitches should agree as functions of ζ even though they differ as functions of Δk . Section 5 tests exactly this.

A back-of-the-envelope magnitude. The row argument also supports a quantitative estimate. For this we assume a simplified setting: constant weights, a continuous uniform orbit, and the one-dimensional geometry of Figure 1. Fix a view and a channel. The channel selects, within the slice at offset z , a line of voxels across the reconstruction — drawn in Figure 3 — and each of those voxels projects along its own ray to the detector. The voxel at horizontal distance d from the source lands at row $z(D_{\text{sdd}}/d)/\Delta_{\text{row}}$, so moving *along the line*, from voxel to voxel at fixed z , the landing row changes at rate $z D_{\text{sdd}}/(d^2 \Delta_{\text{row}})$ per unit of d . The stretch of the line over which the landing row changes by less than one row — the *coherent run* — therefore has length

$$\ell(z) \approx \frac{\Delta_{\text{row}} d^2}{z D_{\text{sdd}}} \quad \text{or, in voxels and row offsets,} \quad \frac{\ell}{\Delta_{xy}} \approx \frac{D_{\text{sid}}/\Delta_{xy}}{\zeta}, \quad (19)$$

where Δ_{xy} is the in-plane voxel pitch and D_{sid} the source-to-rotation-axis (iso) distance — the value of the voxel-dependent d at the volume center: *the coherent run, in voxels, is the source-to-axis distance in voxels divided by ζ* .

What a run does to the gain is a matter of where the squares are taken. The voxels of one run land in the same detector cell, so they contribute to the same entry of $A\mathbf{1}_k$: their contributions are summed *before* the norm in Eq. (4) squares them, and a run of L voxels contributes L^2 (in per-voxel units) to the numerator of the gain. Distinct runs land in distinct cells, whose squared entries simply add: r runs of length L contribute rL^2 , not $(rL)^2$. The denominator squares each voxel’s footprint separately no matter where it lands — that is what the diagonal is — contributing rL . Along one channel the gain is therefore approximately $rL^2/(rL) = L$, the run length itself; and since a run cannot outgrow the reconstruction, it is limited by the in-plane extent, which we take to be D_{rec} , the diameter of the largest circle inscribed in the reconstruction:

$$\hat{g} \sim \min\left(\frac{D_{\text{rec}}}{\Delta_{xy}}, \frac{D_{\text{sid}}/\Delta_{xy}}{\zeta}\right), \quad (20)$$

plus an additive far-field floor from the in-plane footprint overlap of neighboring voxels, which this one-dimensional argument ignores. The estimate thus predicts a gain plateau set by the inscribed diameter, a crossover at $\zeta^* \approx D_{\text{sid}}/D_{\text{rec}}$, and a $1/\zeta$ envelope beyond it. Each landmark has a closed form in the scan geometry. The raw plateau is $D_{\text{rec}}/\Delta_{xy} = n_r$ for a reconstruction of n_r voxels across. The crossover is voxel-free and, for a reconstruction filling a transaxial fan of half-angle γ (so $D_{\text{rec}} = 2D_{\text{sid}} \tan \gamma$), reduces to

$$\zeta^* \approx \frac{D_{\text{sid}}}{D_{\text{rec}}} = \frac{1}{2 \tan \gamma} : \quad (21)$$

the plateau width in rows is set by the fan angle alone — independent of the voxel size, and of D_{sid} and the reconstruction width separately. Widening the reconstruction at a fixed source position narrows the plateau; moving the source back at a fixed field of view widens it.

Everything in the estimate is computable from the scan geometry alone, so it makes falsifiable predictions before any operator is probed: the peak plateau contains no axial quantity, so it should be independent of the slice pitch; the run length depends on the axial geometry only through ζ , so envelopes measured at different pitches should collapse in ζ while the floor — outside the one-dimensional argument — need not; the decay should follow $1/\zeta$ over an additive floor, which once floor-normalized produces log–log slopes shallower than -1 over any finite fitted window; and the numbers can be evaluated per scan. For the widest-cone geometry, $D_{\text{sid}}/\Delta_{xy} = 931$ and $D_{\text{rec}}/\Delta_{xy} = 766$ voxels give a predicted peak self-gain ≈ 770 , a crossover $\zeta^* \approx 1.2$ rows, and a fall to half peak near $2\zeta^* \approx 2.4$ rows — Figure 3 shows the construction, and Figure 4 the resulting predicted profile, annotated with these values. Section 5 confronts each of these predictions with the measurements.

In the continuous constant-weight limit these statements can be organized as properties of the approximately convolutional kernel of $A^\top A$ restricted to a slice [1], with the row sampling supplying the run-length cutoff. The estimate ignores the weights, view nonuniformity, within-slice variation of the Hessian diagonal, and footprint interpolation; it is an order-of-magnitude account, and the measurements remain the arbiter.

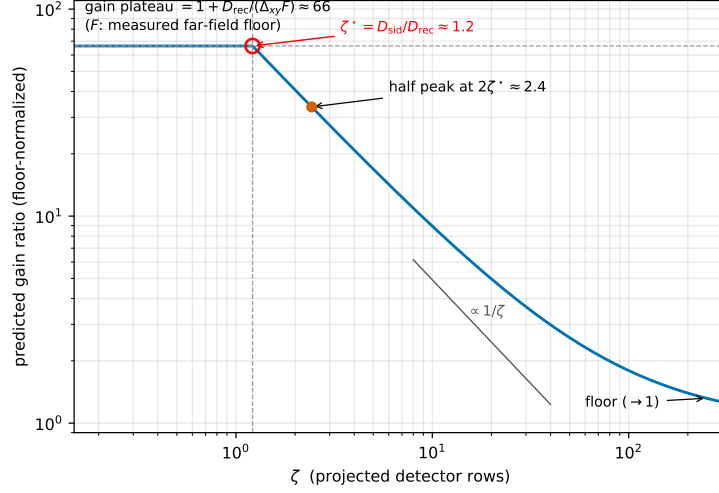


Figure 4: The predicted profile of Eq. (20) for the widest-cone geometry, floor-normalized, with the evaluated values marked: the gain plateau $1 + D_{\text{rec}}/(\Delta_{xy}F) \approx 66$, the crossover $\zeta^* = D_{\text{sid}}/D_{\text{rec}} \approx 1.2$ rows, the fall to half peak at $2\zeta^* \approx 2.4$ rows, the $1/\zeta$ decay, and the approach to the floor. The floor F is the one non-geometric input, taken from the measurement.

5 Measurements

Section 3 ended with three empirical questions — how large g is near the source plane, where the line search actually puts the step size, and whether the single-slice DC family is in fact the non-contracting eigendirection — and Section 4 added a fourth: does the row-offset variable ζ organize the profile across samplings? A fifth question is practical: which scans actually develop the artifact? This section defines one instrument per question and reports the results. [TODO: the eigendirection and drive subsections below]

The probe estimator for the per-slice gain. For a probe slice k , form the lone indicator image $\mathbf{1}_k$ and apply the diagonally preconditioned data operator: one forward projection, a pointwise weighting, one back projection, and a pointwise division by the Hessian diagonal. The statistic — the *self-gain* — is the mean of the response over the probe slice’s own in-support voxels:

$$\hat{g}(k) = \frac{1}{N_k} \mathbf{1}_k^\top \text{diag}(A^\top W A)^{-1} (A^\top W A) \mathbf{1}_k \quad (22)$$

(the $1/\sigma_y^2$ factors cancel). When the Hessian diagonal is constant within the slice, $\hat{g}(k)$ equals the gain $g(k)$ of Eq. (7) exactly; otherwise it is the same ratio with a D -weighted numerator, and we use it as the estimator throughout. Three rules govern the instrument. Probes are *lone* single-slice indicators, one per operator application: probing several slices at once contaminates distant readings through the response’s long one-sided tails (early multi-slice probes inflated the far-field reading roughly twofold before this rule was adopted). Each offset Δk is probed on both sides of k_0 where the volume allows and the sides are averaged. The grid is placed so that k_0 is an integer (the plane centered in one slice). The cost is two projections per probed slice.

Far from the plane the self-gain settles to a roughly constant *floor* — 11.7, 14.2, and 17.3 on the three scans below — reflecting the in-plane coherence every slice shares: neighboring voxels’

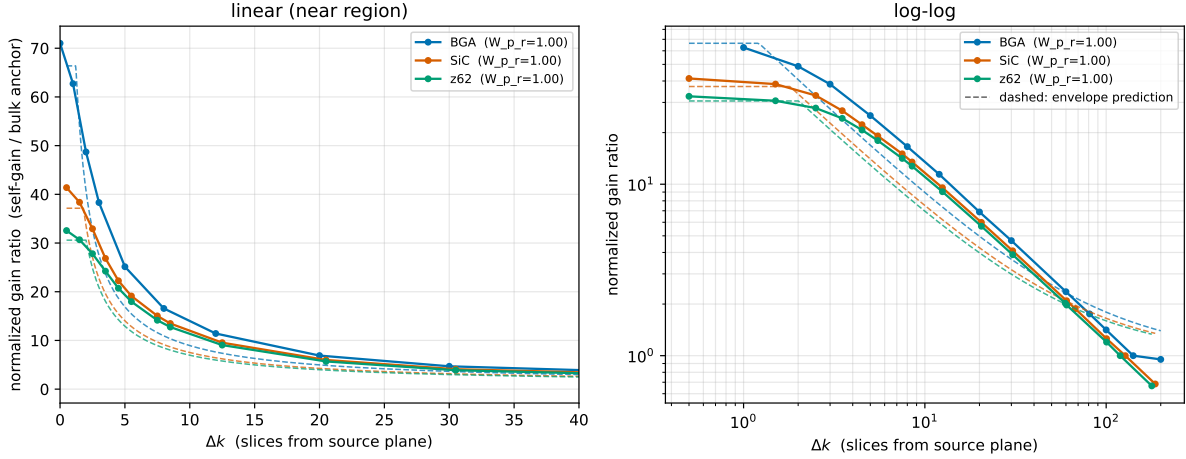


Figure 5: The per-slice gain profile (floor-normalized self-gain, Eq. (22)) versus Δk on three production scans, all at matched pitch ($W_{pr} = 1.00$): linear scale over the near region (left) and log-log over the full range (right). Dashed: the envelope prediction of Section 4 evaluated per scan, with each scan’s measured floor as its one non-geometric input.

scan	D_{sdd} (mm)	D_{sid} (mm)	Δ_{row} (mm)	Δ_{xy} (mm)	n_r	$\tan \gamma$	W_{pr}
BGA	278.5	25.55	0.299	0.0275	766	0.41	1.00
SiC	49.8	22.06	0.054	0.0239	512	0.28	1.00
z62	55.7	50.01	0.054	0.0485	512	0.25	1.00

Table 1: Scan geometries. $\tan \gamma = (n_r/2) \Delta_{xy}/D_{sid}$ is the transaxial fan half-angle at the reconstruction’s edge; the envelope quantities D_{sid}/Δ_{xy} and $D_{rec} = n_r \Delta_{xy}$ follow from these columns.

footprints overlap within a view regardless of axial position. The profiles reported below are floor-normalized, $\hat{g}(k)/\hat{g}_{floor}$, i.e. the plane-specific contrast above the ambient level; the floor itself is taken as the median self-gain over a band of far-from-plane anchor slices away from the volume ends. The remedy of Section 6 is built from this contrast; the role of the absolute floor is taken up there.

The per-slice gain across scans. Figure 5 shows the profile on three scans spanning a range of cone angles, all at matched pitch; Table 1 lists their geometries. The peak contrasts are 71, 41, and 33, ordered by cone angle, and the profiles decay to the floor by $\Delta k \approx 150$ –200. Two observations organize the comparison: the differences between scans concentrate in the near region ($\Delta k \lesssim 8$), and beyond $\Delta k \approx 10$ the three floor-normalized profiles nearly coincide. Across these scans, then, the profile differs mainly by its peak amplitude — similar in shape, in exactly this sense.

The row-offset variable: the pitch test. Equation (18) predicts that the profile is a function of the projected row offset ζ rather than the slice index. The test: re-measure one scan’s profile with the slice pitch scaled by 0.75, 1.00, and 1.50 (the slice count held fixed, so only the pitch — hence W_{pr} and the row span per slice — changes), with the same probe ladder and anchors. Figure 6 and three numbers give the result. First, the peak self-gain is pitch-independent to 0.3%

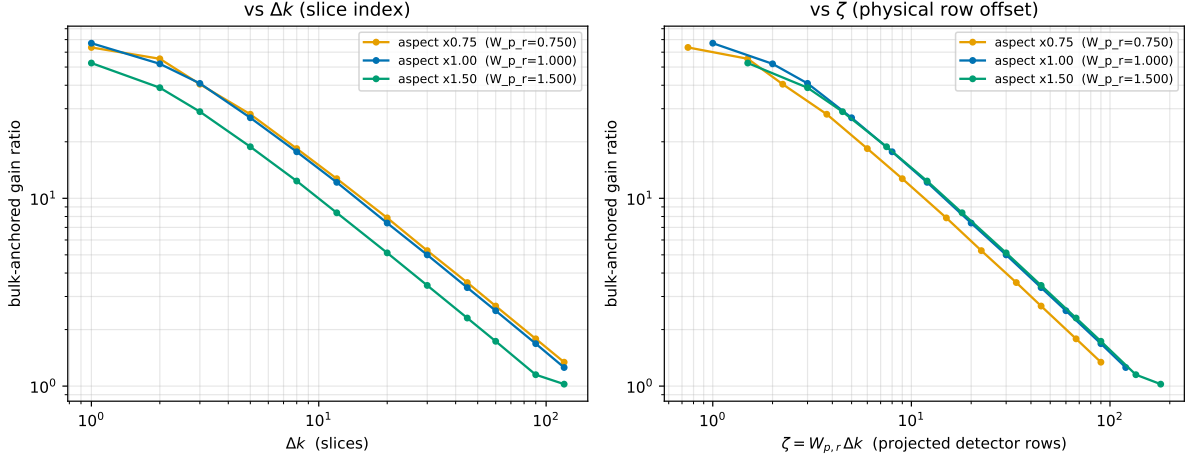


Figure 6: The pitch test: the per-slice gain profile of one scan measured at three slice pitches ($0.75\times$, $1.00\times$, $1.50\times$ the matched pitch; slice count held fixed), plotted against the slice offset Δk (left) and against the projected row offset $\zeta = W_{p,r} \Delta k$ (right).

(832.1, 832.1, 829.4 raw): the $\zeta = 0$ coherence is a property of the plane, not of the sampling — the cleanest single confirmation of the row-coherence mechanism. Second, re-expressed in ζ the two curves with $W_{p,r} \geq 1$ agree within $1.13\times$ over the shared range $\zeta \in [1.5, 90]$ rows, while the finer-than-row curve ($W_{p,r} = 0.75$) runs up to $1.26\times$ low in the tail ($1.29\times$ maximum spread across all three). Third, the anchor-free absolute floor is pitch-dependent and non-monotonic (11.9, 11.0, 14.0), so the unnormalized curves spread wider ($1.51\times$) than the floor-normalized ones. Summary: ζ organizes the profile to first order over this factor-two pitch range — exactly at the peak, approximately (to $\sim 25\%$) in the tail — with the residual dominated by the pitch dependence of the floor and by a one-sided deviation for slices finer than a projected row. A plausible reading of that deviation, stated as interpretation rather than measurement, is that once several slices fit inside one projected row, coherence leaks between adjacent slices and the single-slice framing under-counts it. Finally, at mismatched pitch a grid-commensurability oscillation with period $1/|1 - W_{p,r}|$ slices could ride on the profile; it is finer than this probe ladder, and was neither seen nor excluded at this sampling.

The envelope against the measurements. The estimate of Section 4 made its predictions from geometry alone; each can now be scored. The peak plateau contains no axial quantity, and the measured peak is pitch-independent to 0.3%. The run length depends on the axial geometry only through ζ , and the envelope collapses to first order in ζ while the floor varies with pitch — exactly the split the one-dimensional argument draws. A $1/\zeta$ envelope over an additive floor yields floor-normalized log-log slopes shallower than -1 over any finite window, and the measured profiles show exactly that (power-law fits over the intermediate range give -0.7 to -0.8). Finally, evaluating the envelope per scan — with each scan’s measured floor as its one non-geometric input — predicts both the peak ratios and the half-heights (Table 2):

The dashed curves in Figure 5 are these predicted profiles. The predicted peaks land 6–10% low and the predicted widths 26–36% low — systematic, not scatter — and the ordering across scans, from narrowest to widest, is reproduced. The residual pattern, with the measurement running both taller and wider than any single-length envelope, is the signature of the chord-

scan	peak ratio		half-height (rows)	
	pred.	meas.	pred.	meas.
BGA	66	71	2.4	3.3
SiC	37	41	3.6	5.0
z62	31	33	4.0	6.2

Table 2: The envelope’s per-scan predictions against the measurements. Geometry from Table 1 ($D_{\text{sid}}/\Delta_{xy} = 931, 923, 1031$ voxels and $D_{\text{rec}}/\Delta_{xy} = 766, 512, 512$ voxels for BGA, SiC, z62); each scan’s measured floor is the one non-geometric input.

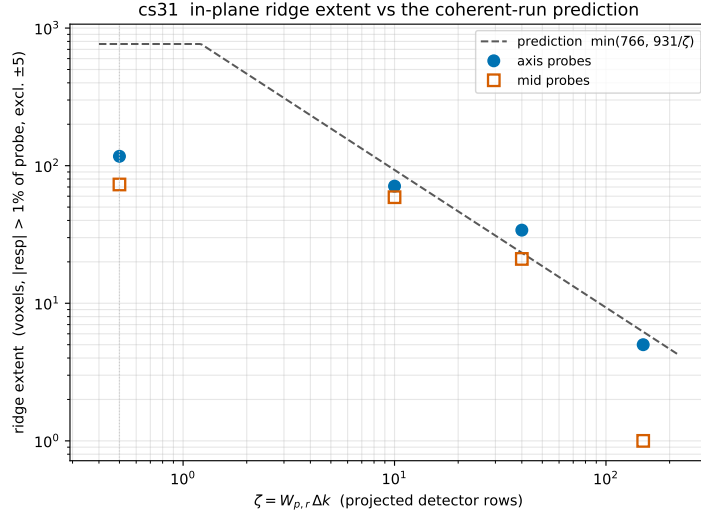


Figure 7: The in-plane coupling extent of the single-voxel rows of Figure 2 (the stretch of the probe’s own slice where the response exceeds 1% of the probe value, threshold fixed in advance), against the coherent-run prediction $L(\zeta) = \min(766, 931/\zeta)$ voxels.

length distribution that the single scale D_{rec} compresses: the longest chords set the peak, while channels with shorter chords remain diameter-limited out to larger ζ and prolong the shoulder. (At the far end of the range the measured ratios dip slightly below 1 — the anchor slices sit inside a still-declining profile — which the floored model cannot follow; that is an artifact of the anchor position, not of the mechanism.)

A sixth check uses a different observable entirely. The single-voxel rows of Figure 2 yield an in-plane coupling extent per probe; at $\Delta k = 10, 40$, and 150 the measured extents are 71, 34, and 5 voxels against the run-length prediction 93, 23, and 6 (Figure 7) — the *column* structure of the operator tracks the same coherent run that the row-sum profiles do. (At $\Delta k = 0$ the statistic reads 117: the band spans the full diameter in the image, but its outer reach falls below the fixed threshold, so the statistic is a conservative lower bound there.) For an argument with no fitted parameters, this is the level of agreement one should hope for, and it says the row-coherence mechanism is not merely qualitative.

The step size, and where the contraction boundary lies. Every quantity so far has described the operator. The criterion of Section 3, however, compares the operator against the

scan	bulk		curvature ratio		α	overshoot αR		
	F	$\bar{p}_h/\bar{f}_h$	R_{bulk}	R_0		bulk	plane	$g_{\text{rel}}(0)$
BGA	11.7	2.3	4.3	284	0.260	1.1	74	71
SiC	14.1	10.6	2.1	40	0.953	2.0	38	41
z62	17.2	7.4	3.0	44	0.914	2.8	40	33

Table 3: The realized step size against the curvature it meets. F is the fh-only self-gain floor the profiles are normalized by; R_{bulk} and R_0 are the curvature ratio of Eq. (23) at the bulk anchors and at the source-plane slice; α is the median line-search step size over the last four of fifteen undamped iterations; and $g_{\text{rel}}(0)$ is the floor-normalized peak gain — the quantity the damping profile of Section 6 takes as its input. Contraction requires $\alpha R < 2$.

iteration: a direction fails to contract when $\alpha\lambda \geq 2$, with $\alpha = 1/R(\tilde{v})$ chosen by the line search. Both sides must be placed on one scale before the measured profile can be read against the criterion, and the profiles above are reported floor-normalized while the criterion is absolute. This measurement supplies the missing conversion.

The instrument is one run per scan, at the configuration that produced the profiles above, so that nothing is compared across cells. It records two things. First, the realized step size: α as the algorithm’s own line search chooses it, logged every iteration (the mean over that iteration’s subsets) through fifteen undamped iterations from the FDK start, and reported as the median over the last four — by which point it has settled. Second, the same lone single-slice probe of Eq. (22), applied at the source-plane slice and at the same bulk anchors that set the floor, but evaluated under the denominator the algorithm actually divides by rather than the data Hessian’s diagonal alone. Writing Δk for the probe’s distance from the plane, the resulting curvature ratio is

$$R(k) = \frac{g(\Delta k) \bar{f}_h(k) + \bar{p}_h(k)}{\bar{f}_h(k) + \bar{p}_h(k)}, \quad (23)$$

the Rayleigh quotient of the in-plane-mean direction on the pencil formed by the full Hessian and the full preconditioner. The prior’s share is taken as the full local curvature $\bar{p}_h$, which is exact for the sparse subset means the iteration actually moves along (Section 6). The run’s per-slice means $\bar{f}_h$, $\bar{p}_h$ come from the same state, and its floor-normalized peaks reproduce those reported above to within rounding (70.9, 41.5, 33.2), confirming that the configuration matches.

Table 3 reports the result, and three readings follow from it.

First, $R(\tilde{v}) = 1/\alpha$ exceeds 1 on all three scans, but by a wide margin only on the widest cone: 3.8, 1.05, 1.09. The line search is therefore not calibrated to isolated voxels, and the threshold in Eq. (15) is not the value 2 that such a calibration would give; it is scan-dependent, and on the worst scan nearly four times larger.

Second, the fh-only floor of 11.7–17.2 is largely an artifact of its normalization. Far from the plane the prior carries most of the per-voxel update denominator — $\bar{p}_h/\bar{f}_h$ runs from 2.3 to 10.6 — so on the scale the algorithm actually steps by, the ambient coherence is $R_{\text{bulk}} = 2.1$ to 4.3, a factor of a few rather than a dozen. Multiplying by the realized step size gives the overshoot the criterion cares about, $\alpha R_{\text{bulk}} = 1.1, 2.0, 2.8$: the volume’s ambient coherent content is stepped at roughly its own Newton length, at or near the contraction boundary 2, and bounded there. That is the answer to the apparent paradox of a floor far above 1. It is worth stating what this

does *not* say: a bulk slice’s DC direction is not an eigendirection of the operator — away from the plane the operator’s rows are broad bowties spanning many slices (Figure 2) — so R_{bulk} is a Rayleigh quotient, hence a lower bound on λ_{max} , and not evidence of a separate non-contracting family in the bulk. In all three scans λ_{max} is certified by the plane, where R is one to two orders of magnitude larger.

Third, and most usefully, the plane’s overshoot αR_0 — 74, 38, 40 — agrees with the floor-normalized peak gain $g_{\text{rel}}(0)$ — 71, 41, 33 — to within 4%, 9%, and 21%. The agreement is not a coincidence but a consequence of the second reading: because $\alpha R_{\text{bulk}} \approx 1$, multiplying by α divides by the ambient level, and the ratio of plane to ambient is what floor normalization computes. The practical consequence is the one the remedy needs: the floor-normalized profile is not a convenient proxy for the overshoot factor but, to within about twenty percent, the overshoot factor itself.

Two limits on how far these numbers should be pushed. The bulk anchors sit inside a profile that is still declining (the same anchor-position effect noted above), so R_{bulk} overstates a true far-field level and the individual anchors spread by a factor of about three. And α varies with subset granularity within an iteration as well as across iterations; the quoted value is the settled late-iteration mean, not a constant of the run. The readings are accordingly good to tens of percent — which is the accuracy the argument requires, since the quantities being separated differ by factors of 30 to 70.

The eigendirection identification. [TODO: the power-method experiment: top eigenmode of the preconditioned normal operator; the in-plane mean share of the mode energy at its peak slice; the mode envelope against the artifact’s axial noise profile.]

Drive versus gain. [TODO: the severity instrument: per-slice update norms; the measured peak-to-floor drive separation between artifact-forming and immune scans; why gain alone does not predict severity.]

6 The remedy

The analysis and the measurements together support a specific and actionable claim: along one identifiable family of directions, and only near the source plane, the step the iteration takes is too long by a factor that can be computed in advance. The remedy is to divide that factor out. Turning that sentence into an algorithm means settling, in order, what the factor is, what it should be measured against, how its two reconstruction-dependent ingredients are estimated, how deep the correction should be allowed to go, and what exactly it multiplies. Two questions of engagement follow: where the source plane is placed on the slice grid, and how the correction stands itself down on data that does not need it.

What the step is actually divided by. Section 3 worked with the data term alone, since that is where the geometry acts and where the large curvature ratio arises. The algorithm’s step, however, divides by $f_h + p_h$: the prior contributes to the denominator as well, and a correction to that step must be referred to the quantity the step actually used. The object to study is therefore the pair

$$(H + H_\rho, \text{diag}(f_h + p_h)), \quad (24)$$

the full Hessian of the cost (1) against the full preconditioner, where $H_\rho = \nabla^2 \rho$ is the prior’s

Hessian. Let v be a unit-norm perturbation supported on slice k and constant on its support. The factor by which the separable model understates the curvature along v is then

$$g_{\text{eff}}(k) = \frac{v^\top (H + H_\rho) v}{v^\top \text{diag}(f_h + p_h) v} = \frac{g(k) \bar{f}_h(k) + v^\top H_\rho v}{\bar{f}_h(k) + \bar{p}_h(k)}, \quad (25)$$

where the denominator is the support mean of the per-voxel curvature, and the data part of the numerator is $g(k) \bar{f}_h(k)$ by the definition of the gain in Eq. (7). Only the prior's contribution $v^\top H_\rho v$ remains to be identified, and it requires some care, because — unlike the data term — it depends on the shape of v 's support.

What the prior contributes depends on which direction is damped. The qGGMRF's curvature is that of a weighted graph: to second order about the current state,

$$v^\top H_\rho v = \sum_{\{i,j\}} b_{ij} (v_i - v_j)^2, \quad (26)$$

the sum running over neighboring voxel pairs $\{i, j\}$ with positive weights b_{ij} , and the per-voxel prior curvature of Eq. (2) being the total weight at a voxel, $p_h(j) = \sum_i b_{ij}$. Written this way, Eq. (26) has an immediate reading for the directions we care about. If v is constant on a set S of voxels and zero elsewhere, then only the pairs that *cross the boundary* of S contribute: a pair with both ends inside S is moved rigidly, its difference does not change, and it drops out.

The two cases in play sit at opposite ends of that statement. For S an entire slice — the direction u_k that Section 3 analyzes — every in-plane neighbor pair lies inside S and cancels, and only the axial pairs cross. The prior then contributes just its axial-neighbor share of $\bar{p}_h(k)$, a small fraction of the whole. But the update the algorithm actually damps is not a whole slice. It is the in-plane mean of one VCD *subset*, and at the granularities where the artifact is driven the subset is a sparse decimation of the pixel grid in which no two selected pixels are neighbors. Every neighbor pair then crosses the boundary of S , Eq. (26) collects the full weight at each selected voxel, and $v^\top H_\rho v = \bar{p}_h(k)$ exactly — the full local prior curvature, not a fraction of it.

The implemented profile uses the full $\bar{p}_h(k)$. By the paragraph above that is exact for the direction actually being damped in the fine-granularity regime, which is the regime in which the artifact is driven; it overstates the prior's share only in the coarse limit, where a subset approaches a whole slice. In that limit the true prior share is smaller, so the true overshoot is larger than Eq. (25) reports and the profile below damps slightly *more* deeply than the exact correction requires. That is the harmless direction for the error to run: it errs toward restoring contraction rather than away from it, and it does so only on the first few, coarsest iterations.

The damping profile. The step along the damped direction is too long by the factor g_{eff} , so multiplying that step by $1/g_{\text{eff}}$ restores the length the true curvature calls for. Substituting the prior share just settled into Eq. (25) gives the per-slice **damping profile**

$$s(k) = \text{clip}_{[0,1]} \left(\frac{\bar{f}_h(k) + \bar{p}_h(k)}{g(\Delta k) \bar{f}_h(k) + \bar{p}_h(k)} \right), \quad s(k) \leftarrow \max(s(k), s_{\min}) \text{ wherever } s(k) < 1,$$

(27)

where $\Delta k = |k - k_0|$ is the slice's distance from the source plane (Eq. (17)) and $s_{\min}$ is a depth floor discussed below. The clip is defensive rather than active: the tabulated gain is floored at

1 before use, so $s \leq 1$ always, and the correction can only ever shorten a step, never lengthen one.

The formula’s limiting cases come out right with no special-casing, which is the main reason to prefer it to a hand-shaped profile:

- **Bulk slices.** Far from the plane $g = 1$, so $s = 1$: no damping at all, exactly and identically.
- **Data-starved end slices.** At the axial ends of the volume few rays pass through, so $\bar{f}_h \rightarrow 0$ and $s \rightarrow 1$. This is correct: where the prior governs the step, the step is not overstated, because the prior’s curvature is represented faithfully by the diagonal.
- **The source plane.** There g is large and s falls toward $1/g$, held above it by whatever share of the per-voxel denominator the prior carries. That share is not small on any measured scan, so the depth the formula actually asks for at the plane differs considerably from one scan to the next even though all are built from one table — the moderation discussed below.

Why the profile is driven by the bulk-relative gain. One ingredient of Eq. (27) has not yet been pinned down: which gain enters as $g(\Delta k)$. The profiles of Section 5 are reported floor-normalized, and it is the floor-normalized value that the formula uses — $g = 1$ in the bulk, by construction. Section 3’s criterion, however, is stated in absolute terms: an isolated voxel has $g = 1$ exactly, and the threshold for non-contraction is $g \geq 2R(\tilde{v})$. Measured absolutely, no slice in any of these volumes has a gain remotely near 1; the floor alone is 11.7 to 17.3. Read literally, the criterion would condemn every slice in the volume, which is plainly not what is observed. The absolute criterion and the relative input must be shown to be the same statement before Eq. (27) can be trusted.

They are, once two conversions are made — and both are measured in Section 5 rather than assumed. The first concerns the threshold, which is not 2. The criterion’s right-hand side is $2R(\tilde{v})$, and $R(\tilde{v}) = 1/\alpha$ is the curvature of the direction the line search is actually serving; the update is a back-projected residual, smooth in plane, and therefore carries real coherent content rather than resembling an isolated voxel. The measured step sizes put $R(\tilde{v})$ well above 1. The second conversion concerns the gain itself. The floor of 11.7–17.3 is measured against the *data* Hessian’s diagonal alone — the right normalization for isolating geometry, and the one Section 4 predicts. The algorithm divides by $f_h + p_h$, and far from the plane the prior carries most of that denominator, so on the algorithm’s own scale the ambient coherent gain is a small number rather than a dozen.

Put the two conversions together and the picture is consistent, and tighter than one might expect: the volume’s ambient coherent directions sit close to the contraction boundary — close enough that the line search is in effect calibrating itself to them — while the source-plane slice stands above that boundary by the floor-normalized peak, a factor of tens. The artifact is localized because only there is the criterion violated by a large factor, and the floor-normalized profile is precisely the measure of that violation.

That settles the arithmetic. A second reason to divide out the excess rather than the absolute level is independent of it, and concerns what the line search is for. The exact line search supplies exactly one global degree of freedom per subset, and it spends that degree of freedom on the ambient level: a curvature error shared by every slice is absorbed into α and never reaches the

committed step. Only departures from the shared level survive to do damage. A remedy aimed at the absolute level would therefore be redoing work the line search has already done — and redoing it specifically on the volume’s smooth, low-frequency content, which is both the slowest-converging part of an MBIR reconstruction and, at the ambient level, the part currently being stepped at very nearly its ideal Newton length. Equation (27) is built instead to be identically inert wherever the iteration is already at the ambient level: $g = 1$ gives $s = 1$ as a property of the formula, not as the result of tuning.

The two curvature scales, and how they are estimated. Equation (27) needs three inputs. Two of them are properties of the particular reconstruction and are computed once, at the start of the run, from its own operators and initial state:

$$\bar{f}_h(k) = \frac{1}{\sigma_y^2} \cdot \text{mean}_{j: f_h(j,k) > 0} [A^\top W A]_{(j,k),(j,k)} \quad (\text{in-support pixels only}), \quad (28)$$

$$\bar{p}_h(k) = \text{mean}_{j \in \mathcal{S}} \left. \frac{\partial^2 \rho}{\partial x_{j,k}^2} \right|_{x=x^{(0)}} \quad (\mathcal{S} \text{ a fixed random subsample of in-support pixels}). \quad (29)$$

Neither is expensive. The Hessian diagonal in Eq. (28) is already formed for the update itself, so $\bar{f}_h$ costs one reduction over an array that exists anyway. For Eq. (29) a few thousand pixel positions (the implementation uses 4096) estimate a slice mean far more accurately than the profile needs. The prior curvature is evaluated at the initial reconstruction $x^{(0)}$, typically the FDK direct reconstruction: the profile is fixed once and not revised, so the state at which it is built should be the state the first updates actually meet.

The third input is the gain shape $g(\Delta k)$, and it is not measured per reconstruction. It is the tabulated profile of Section 5, interpolated log–log between entries and floored at 1:

Δk	0	1	2	3	5	8	12	20	30	60	100	200
g	71.1	62.8	48.7	38.4	25.1	16.6	11.4	6.90	4.69	2.36	1.42	1.0

Table 4: The tabulated gain shape used by Eq. (27): the floor-normalized per-slice gain measured on the widest-cone scan (Section 5), log–log interpolated in use.

No probe is run inside a reconstruction; the probe estimator of Section 5 is an offline calibration instrument, and Table 4 is its output. This makes the cone-angle dependence deliberately *implicit*, and the choice is worth stating plainly. Across the measured scans the shape of the profile is close to shared while the peak amplitude is not (71, 41, 33, ordered by cone angle), so a narrower-cone scan receives a profile built on an amplitude larger than its own and is damped somewhat more than its geometry alone would ask. Two things moderate the consequences. Each reconstruction’s own $\bar{f}_h/\bar{p}_h$ balance enters Eq. (27) and absorbs part of the difference, since a scan whose prior carries more of the update denominator gets a shallower profile from the same table. That effect is substantial rather than cosmetic: evaluated with the per-slice curvature means measured in Section 5, the depth the profile asks for at the source plane is 0.04 on the widest-cone scan but 0.15 and 0.17 on the two milder ones — so most of the amplitude difference the shared table omits is recovered from each reconstruction’s own curvature scales. And where the plane is not driven at all, the trigger below removes the profile entirely rather than moderating it. What the arrangement costs on a mild scan is a measurement, reported in Section 7.

How deep: the floor $s_{\min}$. On the widest-cone scan Eq. (27) asks for a great deal: evaluated with that scan’s own curvature means it returns $s \approx 0.04$ at the plane, a step-shortening of some twenty-five fold (and it would ask for $1/g \approx 0.014$ were the prior’s share of the denominator negligible). Restoring contraction does not require anything close to either figure. Contraction needs only the per-step multiplier $|1 - \alpha\lambda|$ to fall below 1; once the coefficient along the direction is contracting at all, shortening the step further buys progressively less, while continuing to slow the recovery of the legitimate low-frequency content of the slices at the plane — which appears as a broad, slowly clearing transient elevation around the plane. The two effects cross at a knee, and the knee is a measurement rather than a preference: damping deeper than $s_{\min} = 0.1$ saturates the reduction of the artifact while continuing to add transient cost (Section 7). The implementation therefore floors the profile at $s_{\min} = 0.1$ wherever it damps at all, and leaves $s = 1$ exactly wherever it does not.

Rank-one application. The profile is not applied to the whole update. Only the coherent component — the one the analysis indicts — is rescaled:

$$\tilde{\delta}(\cdot, k) = \delta(\cdot, k) + (s(k) - 1) \bar{\delta}(k), \quad \bar{\delta}(k) = \text{mean}_{j \in \text{subset}} \delta(j, k), \quad (30)$$

or equivalently $\tilde{\delta}(\cdot, k) = (I + (s(k) - 1)P) \delta(\cdot, k)$, with P the orthogonal projector, within slice k , onto the vector that is constant on the subset’s pixels. Everything orthogonal to that vector passes through untouched, at full step size.

The restriction is deliberate on both sides. The per-voxel structure of the update is exactly the regime the separable model was built for and handles well — an isolated voxel has $g = 1$ identically — so there is no reason to slow it, and measurement confirms the concern is not hypothetical: damping the whole slice rather than its mean costs real axial structure, while the rank-one form does not (Section 7). Conversely, the coherent component is where the runaway lives. Section 5’s power-method measurement finds the top eigendirection of the preconditioned operator to be this very family, with the in-plane mean carrying essentially all of the direction’s energy at its peak slice. Equation (30) is thus aimed at the smallest subspace that contains the problem.

Why the remedy cannot move the answer. Write $M_k := I + (s(k) - 1)P$ for the operator Eq. (30) applies to slice k . Since P is an orthogonal projector, M_k is symmetric with eigenvalue $s(k)$ on the range of P and 1 on its orthogonal complement; for $s(k) > 0$ — guaranteed by the floor $s_{\min}$ — it is positive definite, and in particular invertible.

That single fact is enough. The subset update is $x^+ = x + \alpha M \delta$ with $\delta = -D_S^{-1} \nabla_S c(x)$, where $D_S \succ 0$ is the curvature diagonal on the subset’s pixels and $\alpha > 0$. Since M is invertible and D_S^{-1} is nonsingular,

$$x^+ = x \iff M \delta = 0 \iff \delta = 0 \iff \nabla_S c(x) = 0. \quad (31)$$

A point is fixed for the iteration exactly when it is fixed for every subset, and the subsets of a partition tile the pixels, so this happens exactly when $\nabla c(x) = 0$. The damped and undamped iterations therefore have precisely the same fixed points, namely the stationary points of the cost (1). Note that s does not appear in the conclusion: the result holds for *any* strictly positive profile, including a badly chosen one. The damping changes the path the iteration takes and nothing whatever about where it is going. (With the positivity constraint enabled, the same

argument runs with stationarity replaced by the corresponding first-order conditions on the feasible set.)

This is the remedy’s central safety property, and it is what makes a step-side intervention the right class of fix for a path-side artifact (Section 3): the reconstruction that the cost defines is untouched, so the damping cannot bias it, blur it, or trade data fit for smoothness. What it can change is how quickly the iteration arrives and in what order the error components clear — the subject of Section 7.

One neighbouring claim should not be read into this, for precision. Positive definiteness of M does not by itself certify that the damped update is a descent direction, because MD_S^{-1} is symmetric only when the curvature diagonal is constant across a slice’s subset pixels. The guarantee above is about fixed points. That the iteration in fact descends — and reaches a data fit no worse than, and in practice better than, the undamped baseline — is a measurement, reported in Section 7.

The grid snap. Section 4 left one loose end. The source plane sits at slice coordinate k_0 of Eq. (17), which need not be an integer, and when it falls between two slices the coherent direction is shared between the straddling pair rather than concentrated in one slice. The remedy is indexed by slices — it reads a per-slice profile and applies a per-slice scale — so the phase of the grid relative to the plane matters to it twice over: it determines how much of the coherent direction any one slice carries, and hence both the depth the profile should apply there and the contrast the trigger below has to work with.

Rather than accommodate every phase, the geometry setup removes the degree of freedom: the volume’s axial offset is adjusted by at most half a slice so that k_0 lands at a slice center. The adjustment is a no-op, bit for bit, when the plane is already centered.

Centering rather than straddling is a deliberate choice, and notably it is not the choice one would make for the undamped iteration, where splitting the coherent direction across two slices halves the visible artifact. Two measured facts decide it. Under damping the two phases give indistinguishable outcomes (Section 7), so nothing is given up; and centering concentrates the coherent direction in a single slice, which leaves the profile one unambiguous slice to damp and gives the trigger a substantially stronger signal to classify.

The decide-once trigger. The damping is armed by default, and that places a burden on it: on a scan whose source plane is never driven it must do nothing, and it must reach that conclusion by itself. Section 5 supplies the statistic, since it is the realized drive — not the gain — that separates the scans which develop the artifact from those which do not.

Through the first two iterations the updater accumulates, alongside the damped update it commits, the *raw* per-slice drive of the undamped update:

$$r_i(k) = \left(\sum_{\text{subsets of iteration } i} \|\delta_{\text{raw}}(\cdot, k)\|_2^2 \right)^{1/2}, \quad i = 0, 1, \quad (32)$$

where δ_{raw} is the subset update before Eq. (30) is applied. At the end of each of those two iterations it forms the plane-to-bulk ratio

$$R_i = \frac{\max_{k \in \text{plane}} r_i(k)}{\text{median}_{k \in \text{bulk}} r_i(k)}, \quad (33)$$

in which “plane” is the slice at k_0 (or the straddling pair on an unsnapped grid) and “bulk” is a reference band at $\Delta k \in [80, 150]$, held clear of the volume ends. After the second reading a single latched decision is made: if $\max(R_0, R_1) < \theta_{\text{off}} = 1.3$, the damping is disarmed for the remainder of the run.

Two features of that design are forced rather than chosen. The statistic must be the raw drive and not the committed one: under damping the committed updates at the plane read a small fraction of the bulk’s, so a trigger reading them would disarm immediately and re-arm on the following iteration — a limit cycle by construction. And the decision must be latched rather than re-evaluated, because even on the raw statistic the damped iteration’s ratio rides well above the immune band for the whole run, so a symmetric on/off threshold placed in that range chatters. Deciding once, on the two iterations before damping has had time to change the picture appreciably, avoids both failure modes; a continued or resumed run reuses the decision rather than re-deriving it.

The threshold itself sits in a measured gap. On the decision statistic $\max(R_0, R_1)$ the immune scan reads 0.93 while the scans that develop the artifact read 1.65 to 1.88 (Section 7), so $\theta_{\text{off}} = 1.3$ has roughly 30% margin either way. The *individual* readings can fall well below that band — on one driven scan R_0 is 0.91 — which is the second reason the rule takes the larger of the two rather than either alone. Two armed iterations on an immune scan are measured indistinguishable from never damping at all.

7 How well it works

The remedy of Section 6 makes several promises, and this section reports the measurements that test them. The evidence is a release battery: every cell runs the shipped defaults — the grid snap and the armed damping — with its expectations written down before the run, and each scan is measured as a matched A/B pair in which the only thing that differs is whether the knob is armed. Five scans are covered, spanning the range that matters: two that develop the artifact strongly, a mild one, one immune to it, a deliberately mismatched-pitch variant, and a thin slab on which the remedy is expected to stand down entirely.

The instrument. The artifact is elevated *noise*, so the statistic has to separate noise from structure. For each slice, the reconstruction is high-pass filtered (the image minus its 7×7 box mean), restricted to voxels inside a central disk and below 30% of the volume’s intensity scale so that structure is excluded, and the standard deviation of what remains is that slice’s **noise profile** $\text{prof}(k)$. Four numbers summarize a profile: the *peak*, its maximum over $\Delta k \leq 15$; the *base*, its median over the same bulk band $\Delta k \in [80, 150]$ that the trigger uses; and the two whole-volume ledgers $\Sigma = \sum_k \text{prof}(k)$ and $\Sigma^2 = \sum_k \text{prof}(k)^2$, which catch any noise the remedy might move elsewhere rather than remove. The data fit is the forward-model RMSE at the last iteration, comparable within a scan but not across scans.

The artifact is removed, at both budgets. Figure 8 (left) shows the result on the widest-cone scan, and it is the clearest single statement the report can make. Undamped at the default budget, the profile carries a narrow spike reaching 11.7 times the local level. Damped, at the same budget, the spike is gone: the peak falls from 0.336 to 0.039, which is 1.13 times the local level — that is, into the noise of the surrounding slices. Extending the budget does not substitute for the remedy. At 50 iterations the undamped run still carries a peak of 0.165, 5.5 times its

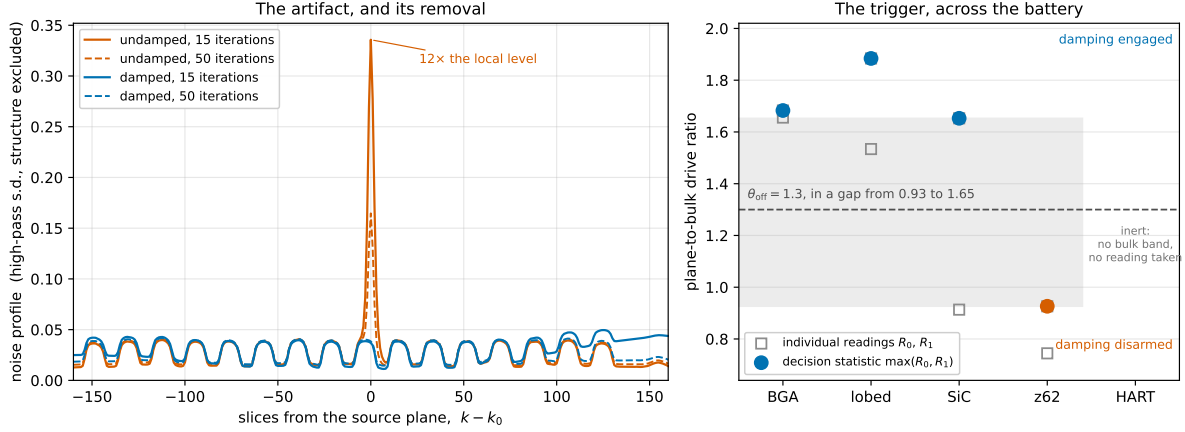


Figure 8: Left: the axial noise profile on the widest-cone scan, undamped versus damped, at the default (15) and extended (50) iteration budgets. The undamped 50-iteration curve is the campaign’s matched baseline; its 15-iteration profile agrees with the battery’s own undamped cell to 7×10^{-5} relative, which is what licenses the shared axes. Right: the trigger’s decision statistic $\max(R_0, R_1)$ across the battery, against the disarm threshold. Open squares are the individual per-iteration readings; note that on one scan the first reading falls *below* the threshold while the second does not.

local level, while the damped run sits at 0.040, or 1.27 times. This is the quantitative form of the claim in Section 3 that the error along the non-contracting direction does not contract: more iterations grind it down slowly, while the remedy removes it outright. Table 5 collects the battery.

What it looks like. Profiles are the right instrument for a ledger, but they are a poor way to see an artifact, so Figures 9 and 10 show the reconstructions themselves, at the default budget, undamped beside damped. Each pair is displayed through an identical window — an A/B with two different windows would show nothing — centered on the background level of that scan’s undamped source-plane slice, with half-width six times its measured bulk noise. The window is tied to the noise scale rather than to the intensity range on purpose: the artifact is a noise excess, so a display scale set by the object’s intensity range would hide it.

Two views are needed because the artifact reads differently in each. In the source-plane slice it is elevated noise, and on the widest-cone scan at the default budget it is overwhelming: the undamped slice is saturated to the point that the object cannot be read at all, while the damped slice taken from the same data at the same iteration shows a row of solder balls and the surrounding structure plainly. In an axial section the same defect becomes a *line* — one slice, spanning the full reconstruction diameter, at k_0 — which is the visual form of the row-coherence argument of Section 4, and it is present in both undamped sections and absent from both damped ones. On the mild scan the effect is smaller in the same places, as the ledgers say it should be.

What the removal costs. Three separate prices could be charged for this, and each was measured.

The first is noise moved rather than removed. It is real, and it is visible in Figure 8 as a slight

Where the damping helps: the trigger engages and the artifact is removed

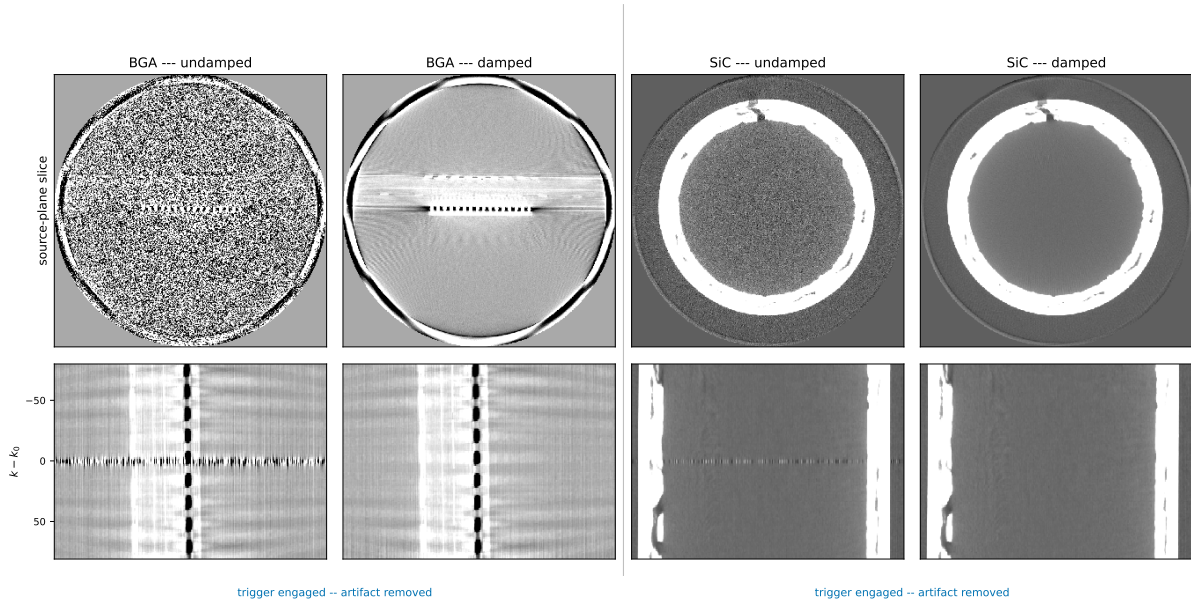


Figure 9: Reconstructions at the default 15-iteration budget on the two scans where the trigger engages, undamped beside damped, each pair through a common window (see text). Top: the source-plane slice, where the artifact is elevated noise. Bottom: an axial section, where the same artifact is a line at the source plane. Nothing differs between the members of a pair except whether the knob was armed — same data, same initialization, same iteration count.

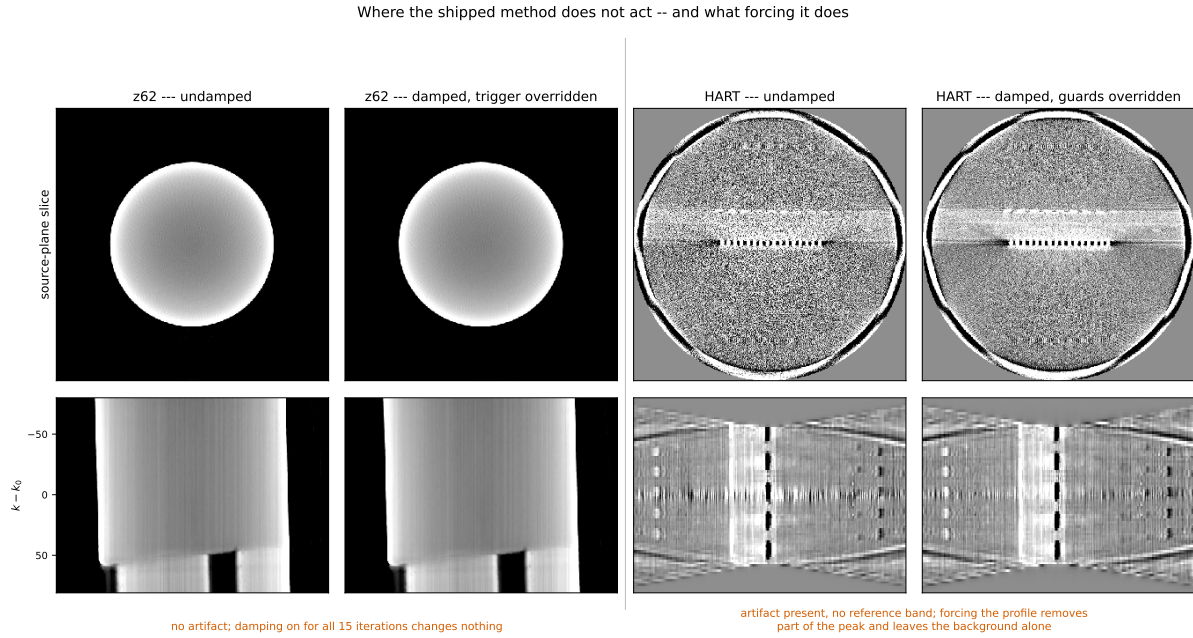


Figure 10: Two scans on which the shipped method does not act, each shown with what it *would* do if forced — the right-hand column of a pair is never the shipped run. Left pair: a scan with no artifact, where neither section carries a line at the source plane, and where damping held on for all fifteen iterations with the trigger overridden changes nothing measurable. Right pair: the thin slab, which does carry a real artifact but is too short to contain the bulk reference band the profile is normalized against, so the method declines; overriding that guard as well applies the tabulated profile anyway. In shipped operation both right-hand columns are replaced by their left-hand neighbors.

scan	knob	iters	peak	peak/base	Σ	Σ^2
BGA	off	15	0.3357	11.73	13.45	0.628
BGA	armed	15	0.0386	1.13	19.33	0.673
BGA	off	50	0.1646	5.50	14.21	0.416
BGA	armed	50	0.0401	1.27	15.26	0.403
SiC	off	15	0.0269	1.46	7.63	0.132
SiC	armed	15	0.0186	1.01	7.65	0.131
z62	off	15	0.0205	0.91	9.26	0.177
z62	armed	15	0.0205	0.92	9.30	0.178
lobed	armed	15	0.0387	1.38	17.42	0.666
HART	armed	15	0.1096	—	2.45	0.111

Table 5: The release battery, recomputed from the saved per-slice profiles with the window definitions above. “lobed” is a mismatched-pitch variant ($W_{pr} = 1.2$); its undamped peak, measured separately, is 0.2834, so the damped value is a $7.3\times$ reduction. HART is the thin slab of Section 8: it carries no bulk band, so no peak-to-base ratio exists and the remedy is inert — its profile is identical to the undamped run.

lift of the damped curves away from the plane: at 15 iterations Σ rises from 13.45 to 19.33. This is a transient rather than a permanent cost — it is the legitimate low-frequency content of those slices converging at the reduced rate the damping imposes — and it washes out with budget: by 50 iterations Σ is 15.26 against the undamped 14.21, while the variance ledger Σ^2 has gone the other way, 0.403 against 0.416. At the extended budget the damped reconstruction is quieter overall as well as free of the artifact.

The second is data fit, and here there is no price at all: the damped run fits the data *better* at both budgets (12.558 against 12.808 at 15 iterations, 11.804 against 11.974 at 50). This is the empirical counterpart of the caveat in Section 6 that positive definiteness of the rank-one operator does not by itself guarantee descent. It also disposes of the obvious suspicion about any noise-reducing intervention — that it buys quiet by fitting the data less well. The opposite happens, for the reason Section 3 gives: the component removed was never signal, so removing it frees the iteration to spend its steps on the rest.

The third is resolution, which is the price a reader should worry about most, since the damped direction is a genuine image degree of freedom. The test is structure-specific: take the sharpest axial edges of the object (the top 2% of $|\partial S/\partial z|$ among structure-adjacent voxels, measured against a converged reference) and compare their amplitude in a band at the plane against the same statistic in a far-from-plane control band. A ratio of 1 means edges at the plane are as sharp as edges anywhere else. The undamped baseline reads 0.95 at 15 iterations and 0.99 at 50; rank-one damping reads 0.96 and 1.00. There is no measurable axial-resolution price at the plane.

That result is not automatic, and the contrast makes the point. The same test applied to damping the *whole* slice update, rather than only its in-plane mean, reads 0.78 at 15 iterations and 0.89 at 50 — a real and partly persistent loss of axial edge amplitude, because a per-slice scalar drags the slice’s structure along with its mean. That is the measurement behind the restriction in Section 6. In the same test the rank-one form does not merely avoid harm: at 15

iterations its structure at the plane is about ten times closer to the converged answer than the undamped baseline’s (0.029 against 0.289 in RMSE), because the noise it removes was what had been impeding convergence there.

The two constants, checked. The profile carries one free constant, the depth floor $s_{\min}$, and the geometry setup carries one convention, the grid phase. Both were fixed by measurement rather than by preference.

Sweeping the damping depth past the floor produces almost no further reduction in the peak — successive depths give 0.0372, 0.0366, 0.0359 — while the data fit steadily degrades, from +0.29% at the knee to +0.9% and +1.6% beyond it, and the noise ledgers worsen with it. The benefit saturates and the cost does not: that is what makes $s_{\min} = 0.1$ a knee rather than a preference.

The grid phase is the one place where the remedy inverts an earlier disposition. Undamped, straddling the plane between two slices is milder than centering it, because the coherent direction is split between the pair. Under damping the two phases are measured indistinguishable — peaks 0.0372 and 0.0372 at 15 iterations, 0.038 at 50 for both — so nothing is given up by centering, and centering buys a single unambiguous slice to damp and a substantially clearer signal for the trigger.

Does it ever make things worse? That is the question to put to any intervention of this kind, and it deserves a direct answer. Inside the envelope the method is built for, no harm case has survived scrutiny — but one appeared to, and the episode is worth telling rather than hiding, because it is this same question asked in its sharpest form.

Early production cells measured the mild scan as *harmed* by damping at the default budget: the peak improved, but the variance ledger rose by 54% and the noise sum by 19%, and it was recorded at the time as a failure. It was not one. The damped and undamped cells being compared had been run under different regularization settings *and* scored with different structure-exclusion masks — two confounds at once, either of them sufficient to manufacture the result. Re-measured with settings and instrument matched, the same comparison is a clean win: peak −31%, variance ledger −0.7%, data fit better (Table 5). The lesson is about measurement rather than about the algorithm, and it is why every cell in this section is a matched A/B: when the effect in question is a few percent on a ledger, the ruler must not be allowed to move.

Two variants genuinely do measure worse, and neither is in the shipped design: damping the whole slice update rather than its in-plane mean costs axial structure, and damping deeper than the floor adds transient cost for no further benefit. Both were reported above. Neither is avoided by luck — the rank-one form and the floor exist precisely because their absence was measured to hurt, which is the pattern worth naming: each guard in the design is there because removing it was tried and cost something.

The thin slab is usually grouped with those two, and on measurement it does not belong there. The method declines to act on such a volume because its profile is normalized against a bulk reference band the volume cannot contain — that is a well-definedness problem, not a measured harm. Forcing the tabulated profile on anyway, with both guards overridden, does not damage the reconstruction: the peak falls from 0.1096 to 0.0758, the background base is unchanged to 0.4%, the variance ledger improves by 7%, and the data fit improves (Figure 10, right pair). It is a partial and untargeted improvement rather than a correction — the tabulated shape is

built for a volume several times longer, and is far too wide for 81 slices — but it is not harm. The honest statement is therefore that the stand-down guard is justified by the profile being undefined there, and not by evidence that acting would hurt.

The trigger. Figure 8 (right) shows the decision statistic across the battery. The three scans that develop the artifact read 1.65, 1.65 and 1.88; the immune scan reads 0.93; the threshold $\theta_{\text{off}} = 1.3$ sits in the gap with about 30% margin either way. Every cell is classified correctly, and the two failure modes the design is meant to avoid do not occur: the immune scan is disarmed after its two measurement iterations and is thereafter indistinguishable from never having been damped at all — peak identical to four decimal places, Σ within 0.4% — and the thin slab, which has no bulk band to reference, is declared inert with the reason recorded rather than guessing.

One detail of the figure is worth dwelling on, because it justifies a design choice that would otherwise look arbitrary. On the mild scan the *first* reading is 0.91, below the threshold, while the second is 1.65, well above it. A trigger reading either iteration alone would have disarmed on a scan that needed damping; taking the larger of the two classifies it correctly. The plane’s drive builds over the first iterations as the subsets refine, so an early low reading is a statement about timing rather than about immunity.

There is a further question here, and the answer qualifies the trigger’s value rather than confirming it. Standing down on an immune scan is only worth doing if damping such a scan would cost something, and that is directly testable: hold the damping on for the whole run with the trigger overridden, and compare. Done on the immune scan, it changes nothing measurable. The peak is unmoved at 0.0205, Σ shifts by +0.07%, Σ^2 by -0.7% and the data fit by -0.1% — all at or below the run-to-run scatter, and all, if anything, in the direction of slight improvement. This is the right-hand column of Figure 10, and it is indistinguishable from the undamped column beside it.

So on the evidence available the trigger is not preventing harm on this scan; it is declining an intervention that would have been harmless. That is a weaker justification than the design assumed, and it should be read together with the coverage limitation in Section 8: the stand-down side of the trigger rests on a single immune scan, and what would strengthen it is not more measurements on that scan but a scan on which unnecessary damping visibly costs something. Until such a case is found, the honest statement is that the trigger is cheap insurance whose premium has been measured and whose payout has not.

Devices, and cost. The trigger reads a ratio of sums over slices, so a reader is right to ask whether it survives being computed across devices in a different order. It does: run at one, two and four GPUs the decisions are identical and the readings agree to four decimal places, and the ledgers match the single-device battery exactly. Cross-layout agreement of the damped reconstructions (3×10^{-5}) is in fact *tighter* than the same comparison with the knob off (1.6×10^{-4}), which is what one would expect from holding down a noise-amplified direction.

The cost is negligible. The profile is built once from quantities the reconstruction already computes, the per-subset work is one mean and one fused multiply on an array already in memory, and no collective is added on the multi-device path. Measured wall clock is within 2% of the undamped run at every device count tested, and +0.8% on one device.

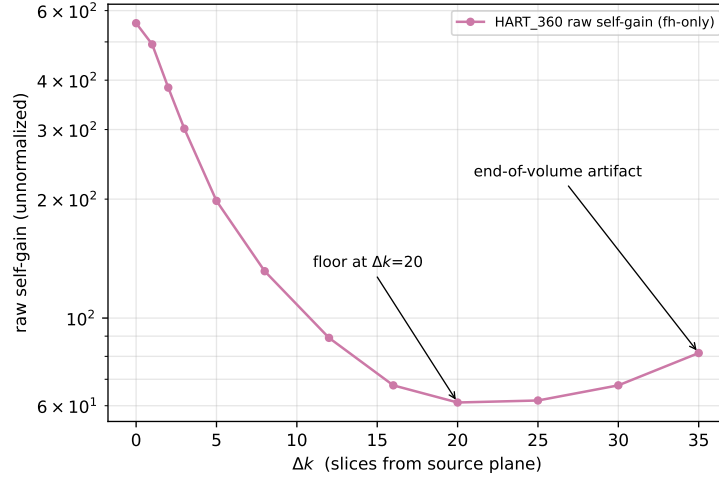


Figure 11: The thin-slab limitation: raw (unnormalized) self-gain on an 81-slice row-cropped scan. The profile has no far-from-plane floor inside the volume — the response turns up past $\Delta k \approx 20$ as the probes approach the opposite volume end — so the floor-referenced profile and trigger have nothing to normalize against.

8 Limitations and possible improvements

Thin volumes. The profile and its trigger are referenced to a far-from-plane floor, and a volume too short to contain one gives the method nothing to normalize against; in that configuration the remedy deliberately stands down (inert, with the reason recorded). The limitation is not hypothetical. Figure 11 shows the probe profile of an 81-slice row-cropped scan: the raw self-gain falls from 558 at the plane to a minimum of 61 at $\Delta k = 20$ and then *rises* — the far probes are already climbing the opposite volume end’s data-starved artifact — so no floor exists inside the window. The peak-to-minimum contrast (≈ 9) is a lower bound; the scan’s true contrast is unmeasurable from this slab.

Mismatched slice/row pitch. At $W_{pr} \neq 1$ the profile in slice-index space differs from the matched-pitch profile (Section 5), a fixed slice-indexed profile mis-shapes accordingly, and the possible grid-commensurability comb remains unresolved at the probe sampling used. The production geometry pipeline chooses matched pitch, so this regime is reached only by manual override — but it is a stated boundary of the profile’s shape claim. [TODO: refine after the remedy section lands]

Validation coverage. [TODO: the immune (stand-down) side of the trigger validation rests on fewer scans than the driven side; state the coverage and what a held-out immune scan would add.]

Improvement directions. Three directions, stated as directions rather than designs.

First, the damping generalizes naturally in frequency. The coherent ridge of length $L(\zeta)$ is, viewed in the in-plane frequency domain, an enhancement confined to $|f| \lesssim 1/L(\zeta)$. At the source plane $L \approx D_{\text{rec}}$, so the enhancement occupies essentially one in-plane Fourier bin — which is why damping the DC bin alone, as Eq. (30) does, suffices there. Off the plane the ridge

shortens and the enhancement correspondingly widens, which suggests a *rank-few* extension: damp the first few in-plane bins of each slice, with per-bin factors measured exactly as the present profile is, by applying the probe estimator of Eq. (22) to low-frequency sinusoid patterns in place of indicator slices. Such an extension keeps the subset structure, and it keeps the safety argument of Section 6 with it: that proof uses only the invertibility of the operator applied to the update, so it covers any collection of positive per-bin factors, whether or not the bins are exactly orthogonal on the decimated subset. It reduces to the present remedy when one bin suffices. Whether anything beyond the DC bin is worth correcting is one measurement away, since the single-voxel rows of Figure 2 are empirical in-plane kernels whose transforms give the per-bin gains directly.

Second, and more ambitiously, the same picture invites a full in-plane Fourier preconditioner: model the in-plane operator as a slice-dependent convolution and invert its symbol. This is a well-established idea in tomography — frequency-domain preconditioning filters designed from the system response go back to Clinthorne et al. [2], who also make the point that such preconditioning changes the convergence behaviour without changing the solution the iteration reaches. The prize here is larger than the artifact. Section 5 shows that the diagonal misrepresents coherent in-plane directions on *every* slice and not only at the plane — away from the plane the ratio R_{bulk} is still a factor of a few — and one global step size cannot serve both ends of a spectrum that wide. The line search must compromise, and the measured α records which compromise it strikes: on the widest-cone scan $\alpha \approx 0.26$, so the directions the separable model represents exactly are advanced at about a quarter of their Newton length; on the milder scans α sits near 1 instead, and it is the ambient coherent content that is left at the edge of contraction. A preconditioner that flattened the in-plane spectrum would remove that compromise rather than manage it. The price is a genuine change of algorithm: a full-plane preconditioner mixes voxels outside the active subset, so the coordinate-descent structure and its line search would have to be rebuilt, and the convolution model degrades where the weights and the support edge make the operator shift-variant in plane — the difficulty that motivated the shift-variant preconditioners of Fessler and Booth [3]. It belongs to the same conversation as multi-resolution schemes, which attack the same low-frequency sluggishness from the scale side rather than the frequency side.

Third, the remedy extends in scope: damping profiles shaped by each reconstruction’s own measured drive rather than by a fixed table, and an extension of the source-plane analysis to helical trajectories, where the coherent plane sweeps axially through the volume rather than standing still within it.

A Reproducibility

[TODO: map each figure and table to its generating script and data location (the campaign’s experiment scripts and depot artifacts), keeping campaign-internal identifiers out of the report body.]

References

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